Operational Expectations of Software Systems

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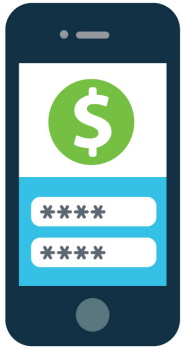
Objectives

Operational expectations of software systems

Explain the durability and changeability expectations of software systems.

Operational for Longer Period of Time

Financial Application



Millions
of lines of code

Modern Car

6-10 Million
lines of code

F-35 Fighter Jet

10 Million
lines of code

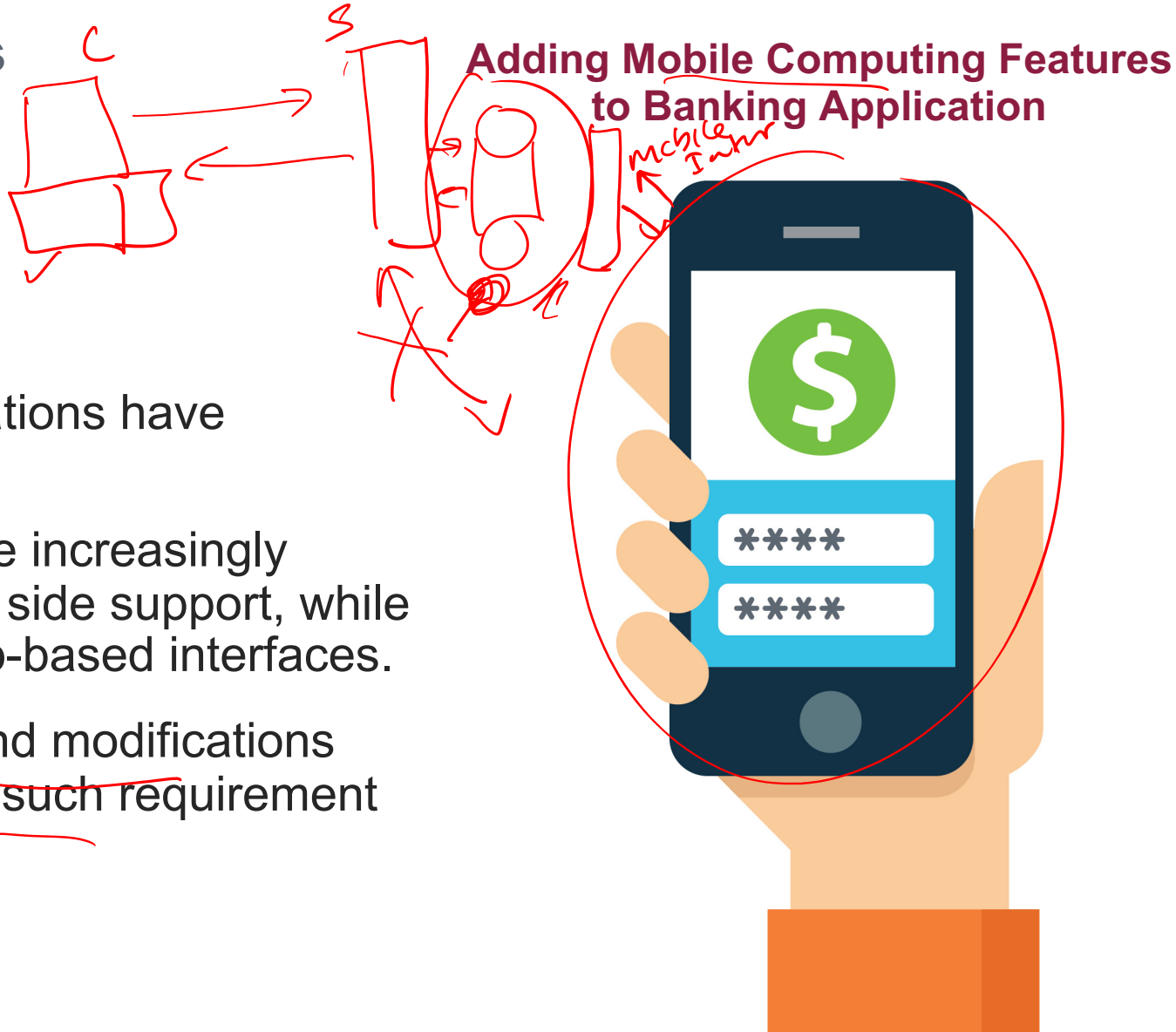
Durability expectations are high for these software systems

Amenable to Changes Due to New Technological Advancements

Software undergoes changes due to technology changes

Example

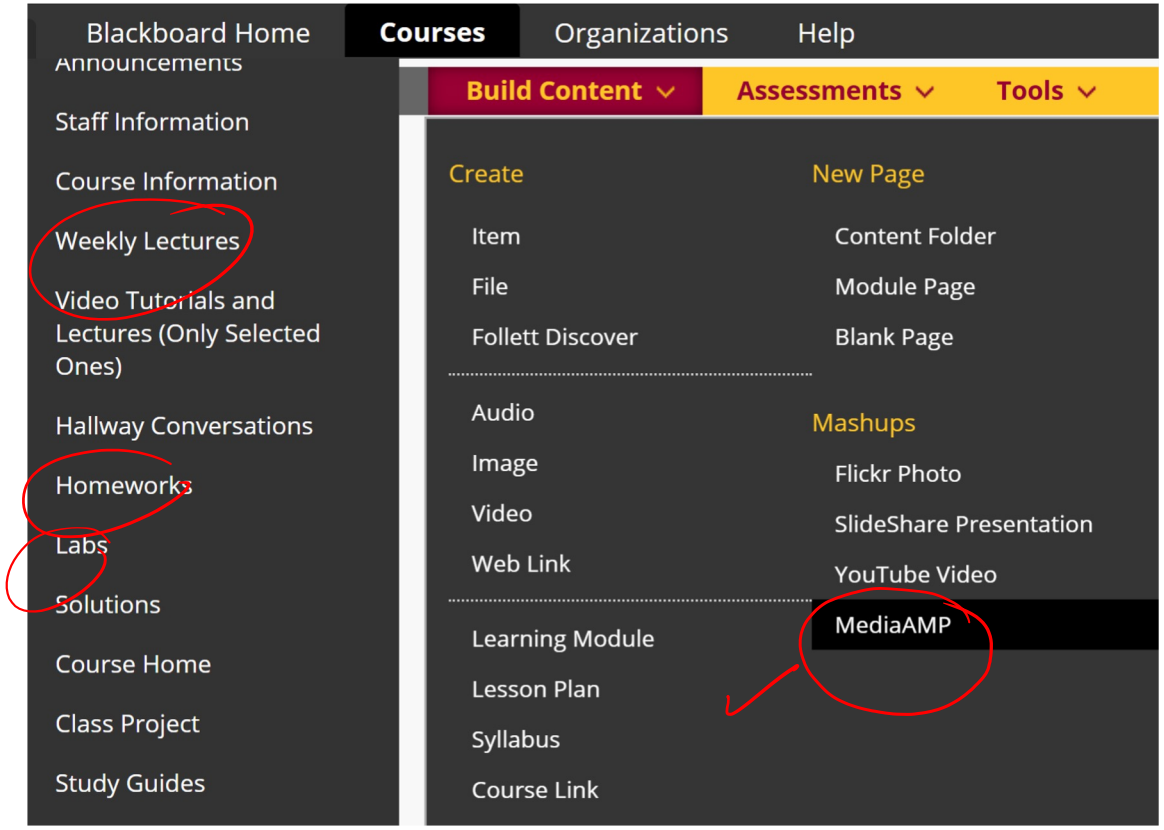
- Today's banking applications
  - Conventional banking applications have web-based interfaces.
  - Now, banking applications are increasingly moving towards mobile client side support, while maintaining conventional web-based interfaces.
  - **Question:** How much backend modifications are needed to accommodate such requirement changes?



Changes Due to Functionality Requirements

- Blackboard, an online learning management system, supports various different features (e.g., gradebook, course material management, email communication, etc.) for course management.
- The flipped-classroom is a new approach to learning where students prepare for class prior to the class meeting and during class time, they participate in more hands-on and problem- solving activities.
- To support the flipped classroom model, new functionalities were introduced on Blackboard (e.g., video/audio embedding features, auto-graded assessment creation, etc.)
- Such functionality changes need new modules to be added to the system.
- How much backend modifications are needed to accommodate such requirement changes?

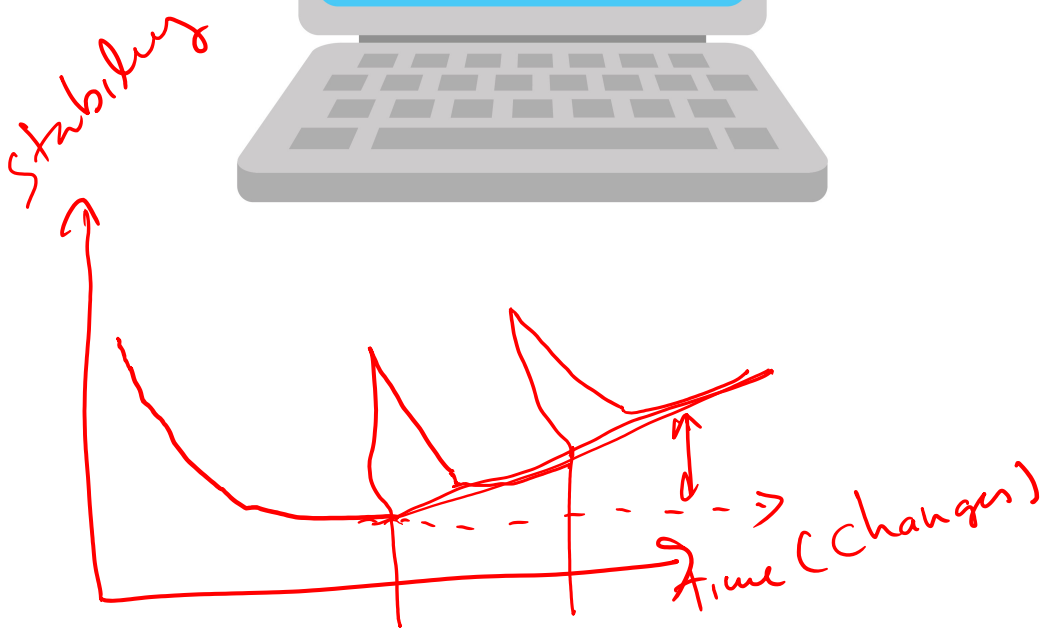
Software Changes Due to New Functionality



Blackboard System Example:
New video/audio features were added to Blackboard

What Happens When Software Undergoes Changes

- **As software changes, the stability of the software system may be affected.**
- **Stability may be related to:**
  - Number of software bugs and severity of those bugs
  - Other quality factors such as modularity, maintainability, reliability …etc
- **The objective is to accommodate changes with minimum or no (sometimes unlikely) affect to the stability of the software.**



Summary

- **Software is becoming (if it has not already) a day-to-day commodity**
- **Two key expectations**
  - Needs to be operational for a longer duration
  - Undergo changes during the life-cycle of the software
- **Key question**
  - How well a software system can fulfil its operational expectations?
- **Answer**
  - It depends - it is dependent on how well the software has been designed to accommodate changes.